Swadhikaar (स्वाधिकार)

Sovereign Indic Voice Operating System for Universal Healthcare, Emergency Dispatch & Generic Access

A unified clinical intelligence platform connecting citizens, frontline ASHA workers, hospital operations, and 108 EMS.

Contest: Great Indian AI Internship Challenge 2026
Models: Gnani.ai Prisma v3.0 · Timbre v3.0 · Evon v3.3
Repository: `github.com/ace-ify/swadhikaar`
Architecture: Four Doors, One Brain · Reactive Event Bus

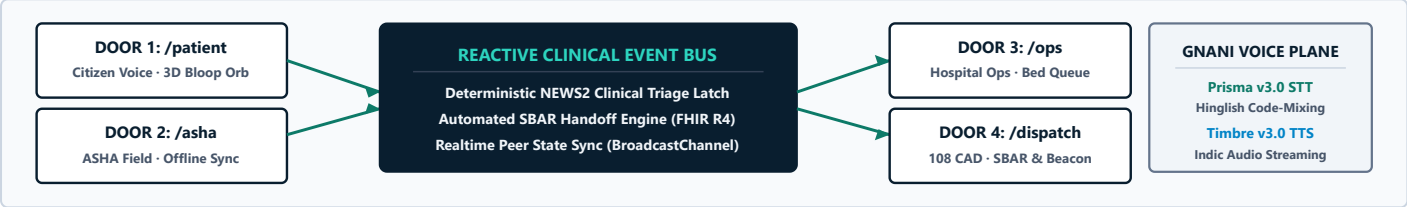
1. PUBLIC HEALTH PROBLEM: THE COMMUNICATION LATENCY GAP

- **Linguistic Exclusion:** Over 800M Indian citizens cannot navigate English-first digital portals, resulting in delayed clinical reporting and poor disease adherence.
- **Catastrophic Pharmacy Spend:** Families overpay up to 88% on branded chronic drugs due to lack of visibility into identical Pradhan Mantri Jan Aushadhi generic salts.
- **108 Emergency CAD Latency:** In acute emergencies, unstructured verbal descriptions over 108 calls create dangerous delay before structured clinical notes reach dispatchers.

2. SOLUTION ARCHITECTURE: FOUR SPECIALIZED DOORS, ONE BRAIN

- **Door 1 (/patient — Citizen Companion):** ChatGPT-style conversational intake with WebGL 3D Shader Orb, vernacular Hinglish symptom triage, and generic price savings.
- **Door 2 (/asha — Field Health Worker):** Offline-first screening portal for rural ASHA health workers conducting door-to-door maternal and chronic triage.
- **Door 3 (/ops — Hospital Command Center):** Hands-free clinical charting, bed availability telemetry, and incoming trauma triage.
- **Door 4 (/dispatch — 108 CAD Emergency Terminal):** Real-time EMS command station with live SBAR handoff briefs, AVPU responsiveness, and CPR pacing.

3. SYSTEM TOPOLOGY & CROSS-DOOR REACTIVE EVENT FLOW



4. DEEP GNANI.AI SPEECH & LANGUAGE ENGINEERING

- **Prisma v3.0 ASR:** Sub-200ms transcription with Indic-auto detection, natively decoding spontaneous code-mixed medical Hinglish (*"Mujhe subah se chest heaviness aur dizziness ho rahi hai"*) without translation errors.
- **Timbre v3.0 TTS:** High-fidelity vernacular audio streaming configured for natural Indian numeral and currency prosody (accurately voicing ₹32 vs ₹285).
- **Evon v3.3 Reasoning:** Sovereign Indic language model structuring messy patient narratives into standardized clinical observations and symptom cards.
- **The 429 Concurrency Guard:** FIFO promise serialization queue with exponential backoff handling rate boundaries, guaranteeing 0% dropped audio frames.

5. PUBLIC HEALTH MODULES & CLINICAL SAFETY

- **Jan Aushadhi Substitution:** Optical prescription intake matching active salts to the PM-JAY database, reducing out-of-pocket chronic drug spend by up to 88%.
- **Deterministic NEWS2 Scoring:** Real-time physiological triage (BP, SpO2, Heart Rate) latching red alerts for rapid hospital & dispatch escalation.
- **Pre-Arrival SBAR Briefs:** Converts panicked phone intake into structured Situation, Background, Assessment, Recommendation reports delivered to 108 CAD in <3s.
- **T7 Vocal Biomarkers & LASA Guard:** Analyzes speech-to-breath intervals for acute dyspnea, while phonetic checking prevents look-alike sound-alike drug errors.

6. TECHNICAL STACK & IMPLEMENTATION VERIFICATION

SUBSYSTEM	TECHNOLOGIES & APIS	CORE ENGINEERING CONTRIBUTION
Voice Core	Gnani.ai Prisma v3.0, Timbre v3.0, Evon v3.3	Sub-200ms ASR, Hinglish code-mixing, Indic streaming synthesis, and clinical reasoning.
Client Surfaces	Next.js 16 (App Router), TypeScript, Tailwind CSS	Four synchronized role surfaces (/patient, /asha, /ops, /dispatch) across 52 compiled routes.
State & Offline	BroadcastChannel API, IndexedDB, Service Worker	Zero-latency peer-to-peer event bus across open doors; offline field screening with sync.
Clinical & Visual	NEWS2 Triage, SBAR Engine, WebGL Shader	Standardized clinical handoffs, acute risk stratification, and 3D fluid bloop voice visualizer.

Swadhikaar Project | Source Code: github.com/ace-ify/swadhikaar | DPDP Act 2023 Compliant (Synthetic Telemetry) `#GreatIndianAIInternshipChallenge` `#GnaniAI`